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Education & Work Experience

- 2025 - pres. Research Scientist, NASA Ames Research Center contracted via the SETI Institute, Moffett Field, CA.
- 2023 - 2025 NASA Postdoctoral Program Fellow, Space Science and Astrobiology Division, NASA Ames Research Center, Moffett Field, CA. Advisor: Natasha Batalha.
- 2017 - 2023 PhD Graduate Student, Dual-title Earth and Space Science and Astrobiology, University of Washington, Seattle, WA. Advisor: David Catling.
- 2012 - 2016 B.S., Physics major, University of Oregon Honors College, Eugene, OR.

Awards & Honors

- 2023 The David A. Johnston Award for Research Excellence, Earth and Space Science Department, University of Washington.
- 2023 NASA Postdoctoral Program Fellowship recipient
- 2023 Award for best graphic in presentation, Earth and Space Science Gala, University of Washington.
- 2017 Department of Earth and Space Sciences Top Scholar Award, University of Washington
- 2016 Undergraduate Research Award, University of Oregon Physics department
- 2016 Undergraduate Honors Thesis: Passed with Highest Distinction, University of Oregon Honors College

Mentoring, Teaching & Outreach

- 2024 - pres. Informally mentoring and collaborating with Conner Metz, a U-M graduate student, studying the photochemistry of sulfur on early Mars.
- 2024 - pres. Informally mentoring and collaborating with Sander Somers, a UMD graduate student, studying photochemical haze production in gas rich exoplanet atmospheres.
- 2024 - pres. Informally mentoring and collaborating with Samantha Gilbert, a UW graduate student, studying the coupled atmosphere-interior evolution of habitable rocky exoplanets.
- 2024 - pres. Informally mentoring and collaborating with John Herring, an ISU graduate student, studying the evolution of nitrogen on early Earth and rocky exoplanets.
- 2025 - pres. Mentoring Jaden Lopez, an undergraduate student at UCSC, through the Lamat Program. Jaden is studying gas opacities for the use of reflected light exoplanet spectroscopy.
- 2025 Mentored Elizabeth Pawelka, a post-baccalaureate researcher, through the NASA OSTEM program. Elizabeth studied sub-Neptune atmospheres in the context of the Habitable Worlds Observatory.

- 2024 - 2025 Informally mentored Sagnick Mukherjee, a former UCSC graduate student, on a first author article that studies photochemistry in hot gas giant atmospheres.
- 2024 - 2025 Informally mentored Jea Adams Redai, a Harvard graduate student, on a first author article publishing JWST observations of the rocky exoplanet GJ 357 b.
- 2024 Mentored Zoe Lu and Divya Bhamidipati, both high school students, through the NASA-Chabot internship program.
- 2022 - 2023 “Rockin’ Out” volunteer. Rockin’ Out is a K-12 volunteer-based outreach program at the Department of Earth and Space Sciences at the University of Washington.
- 2022 Presented at Astronomy on Tap, Seattle: “How Life Started and Where The Universe Will Go”
- 2019 Graduate Teaching Assistant: Intro. Astrobiology (ASTBIO 115; Winter), University of Washington.
- 2018 Teaching Assistant: Intro. Geology (ESS 101; Winter), University of Washington.

Peer-Reviewed Publications

31 total, 7 first-author. Google Scholar: [Linked here](#). * indicates mentored graduate student.

Under Review

- 2025 Joshua Krissansen-Totton, ..., **Nicholas Wogan** et al. (2025). Wavelength Requirements for Life Detection via Reflected Light Spectroscopy of Rocky Exoplanets. Under Review, *Astrobiology*. DOI: [10.48550/arXiv.2507.14771](#).
- 2025 Jegug Ih, ..., **Nicholas Wogan** et al. (2025). Do Rocky Planets around M Stars Have Atmospheres? A Statistical Approach to the Cosmic Shoreline. Under Review, *AAS Journals*. DOI: [10.48550/arXiv.2508.08253](#).

Accepted or Published

- 2025 Sukrit Ranjan, **Nicholas Wogan** et al. (2025). The Photochemical Plausibility of Warm Exo-Titans Orbiting M-Dwarf Stars. Accepted, *The Astrophysical Journal*. DOI: [10.48550/arXiv.2509.10611](#).
- 2025 Amélie Gressier, Natasha Batalha, **Nicholas Wogan** et al. (2025). JWST-TST DREAMS: Sulfur dioxide in the atmosphere of the Neptune-mass planet HAT-P-26 b from NIRSpec G395H transmission spectroscopy. Accepted, *Astronomical Journal*. DOI: [10.48550/arXiv.2509.16082](#).
- 2025 **Nicholas Wogan**, Natasha Batalha, Kevin Zahnle, Joshua Krissansen-Totton, David Catling, Eric Wolf, Tyler Robinson, Victoria Meadows, Giada Arney and Shawn Domagal-Goldman (2025). The Open-Source Photochem Code: A General Chemical and Climate Model for Interpreting (Exo)Planet Observations. Accepted, *The Planetary Science Journal*. DOI: [10.48550/arXiv.2509.25578](#).
- 2025 Stephen Schmidt, Ryan MacDonald, ..., **Nicholas Wogan** et al. (2025). A Comprehensive Reanalysis of K2-18 b’s JWST NIRISS+NIRSpec Transmission Spectrum. Accepted, *The Astronomical Journal*. DOI: [10.48550/arXiv.2501.18477](#).
- 2025 Jea Adams Redai*, **Nicholas Wogan** et al. (2025). JWST COMPASS: A NIRSpec G395H Transmission Spectrum of the Super-Earth GJ 357 b. *The Astronomical Journal*. DOI: [10.3847/1538-3881/adee92](#).

- 2025 Trent Thomas*, ..., **Nicholas Wogan** et al. (2025). Statistical Geochemical Constraints on Present-day Water Outgassing as a Source of Secondary Atmospheres on the TRAPPIST-1 Exoplanets. *The Planetary Science Journal*. DOI: [10.3847/PSJ/add261](https://doi.org/10.3847/PSJ/add261).
- 2025 Sagnick Mukherjee*, Jonathan Fortney, **Nicholas Wogan**, David Sing, and Kazumasa Ohno (2025). Effects of Planetary Parameters on Disequilibrium Chemistry in Irradiated Planetary Atmospheres: From Gas Giants to Sub-Neptunes. *The Astrophysical Journal*. DOI: [10.3847/1538-4357/adc7b3](https://doi.org/10.3847/1538-4357/adc7b3).
- 2025 Kenneth Gordon, Theodora Karalidi, Kimberly Bott, **Nicholas Wogan** et al. (2025). Polarized Signatures of the Earth Through Time: An Outlook for the Habitable Worlds Observatory. *The Astrophysical Journal*. DOI: [10.3847/1538-4357/adc09c](https://doi.org/10.3847/1538-4357/adc09c).
- 2025 Johanna Teske, Natasha Batalha, Nicole Wallack, James Kirk, **Nicholas Wogan** et al. (2025). JWST COMPASS: NIRSpec/G395H Transmission Observations of TOI-776 c, a 2 Rearth M Dwarf Planet. *The Astronomical Journal*. DOI: [10.3847/1538-3881/adb975](https://doi.org/10.3847/1538-3881/adb975).
- 2025 Lili Alderson, Sarah Moran, Nicole Wallack, Natasha Batalha, **Nicholas Wogan** et al. (2025). JWST COMPASS: NIRSpec/G395H Transmission Observations of the Super-Earth TOI-776 b. *The Astronomical Journal*. DOI: [10.3847/1538-3881/adad64](https://doi.org/10.3847/1538-3881/adad64).
- 2024 Nicholas Scarsdale, **Nicholas Wogan** et al. (2024). JWST COMPASS: The 3-5 μ m Transmission Spectrum of the Super-Earth L 98-59 c. *The Astronomical Journal*. DOI: [10.3847/1538-3881/ad73cf](https://doi.org/10.3847/1538-3881/ad73cf).
- 2024 Munazza Alam, Peter Gao, Jea Adams Redai, Nicole Wallack, **Nicholas Wogan** et al. (2024). JWST COMPASS: The First Near- to Mid-infrared Transmission Spectrum of the Hot Super-Earth L 168-9 b. *The Astronomical Journal*. DOI: [10.3847/1538-3881/ad8eb5](https://doi.org/10.3847/1538-3881/ad8eb5).
- 2024 **Nicholas Wogan**, David Catling, and Kevin Zahnle (2024). Timing and likelihood of the origin of life derived from post-impact highly reducing atmospheres. *Astrobiology*. DOI: [10.1089/ast.2023.0128](https://doi.org/10.1089/ast.2023.0128).
- 2024 Joshua Krissansen-Totton, **Nicholas Wogan**, Maggie Thompson and Jonathan Fortney (2024). The erosion of large primary atmospheres typically leaves behind substantial secondary atmospheres on temperate rocky planets. *Nature Communications*. DOI: [10.1038/s41467-024-52642-6](https://doi.org/10.1038/s41467-024-52642-6).
- 2024 Nicole Wallack, Natasha Batalha, ..., **Nicholas Wogan** et al. (2024). JWST COMPASS: A NIRSpec/G395H Transmission Spectrum of the Sub-Neptune TOI-836c. *The Astronomical Journal*. DOI: [10.3847/1538-3881/ad3917](https://doi.org/10.3847/1538-3881/ad3917).
- 2024 Lili Alderson, Natasha Batalha, ..., **Nicholas Wogan** et al. (2024). JWST COMPASS: NIRSpec/G395H Transmission Observations of the Super-Earth TOI-836b. *The Astronomical Journal*. DOI: [10.3847/1538-3881/ad32c9](https://doi.org/10.3847/1538-3881/ad32c9).
- 2024 Shang-Min Tsai, Hamish Innes, **Nicholas Wogan**, Edward Schwieterman (2024). Biogenic Sulfur Gases as Biosignatures on Temperate Sub-Neptune Waterworlds. *Astrophysical Journal Letters*. DOI: [10.3847/2041-8213/ad3801](https://doi.org/10.3847/2041-8213/ad3801).

- 2024 **Nicholas Wogan**, Natasha Batalha, Kevin Zahnle, Joshua Krissansen-Totton, Shang-Min Tsai, and Renyu Hu (2024). JWST observations of K2-18b can be explained by a gas-rich mini-Neptune with no habitable surface. *Astrophysical Journal Letters*. DOI: [10.3847/2041-8213/ad2616](https://doi.org/10.3847/2041-8213/ad2616).
- 2024 Zoe Todd, **Nicholas Wogan**, and David Catling (2024). Favorable Environments for the Formation of Ferrocyanide, a Potentially Critical Reagent for Origins of Life. *ACS Earth and Space Chemistry*. DOI: [10.1021/acsearthspacechem.3c00213](https://doi.org/10.1021/acsearthspacechem.3c00213).
- 2024 Amber Young, Tyler Robinson, Joshua Krissansen-Totton, Edward Schwieterman, **Nicholas Wogan** et al. (2024). Inferring chemical disequilibrium biosignatures for Proterozoic Earth-like exoplanets. *Nature Astronomy*. DOI: [10.1038/s41550-023-02145-z](https://doi.org/10.1038/s41550-023-02145-z).
- 2023 **Nicholas Wogan**, David Catling, Kevin Zahnle, and Roxana Lupu (2023). Origin of life molecules in the atmosphere after big impacts on the early Earth. *Planetary Science Journal*. DOI: [10.3847/PSJ/aced83](https://doi.org/10.3847/PSJ/aced83).
- 2023 Zachary Cohen, Zoe Todd, **Nicholas Wogan**, Roy Black, Sarah Keller, and David Catling (2023). Plausible sources of membrane-forming fatty acids on the early Earth: a review of the literature and an estimation of amounts. *ACS Earth and Space Chemistry*. DOI: [10.1021/acsearthspacechem.2c00168](https://doi.org/10.1021/acsearthspacechem.2c00168).
- 2022 **Nicholas Wogan**, David Catling, Kevin Zahnle, and Mark Claire (2022). Rapid timescale for an oxic transition during the Great Oxidation Event and the instability of low atmospheric O₂. *Proceedings of the National Academy of Sciences*. DOI: [10.1073/pnas.2205618119](https://doi.org/10.1073/pnas.2205618119).
- 2022 Maggie Thompson, Joshua Krissansen-Totton, **Nicholas Wogan**, Myriam Telus, and Jonathan Fortney (2022). The case and context for atmospheric methane as an exoplanet biosignature. *Proceedings of the National Academy of Sciences*. DOI: [10.1073/pnas.2117933119](https://doi.org/10.1073/pnas.2117933119).
- 2021 Joshua Krissansen-Totton, Max Galloway, **Nicholas Wogan**, Jasmeet Dhaliwal, and Jonathan Fortney (2021). Waterworlds probably do not experience magmatic outgassing. *The Astrophysical Journal*. DOI: [10.3847/1538-4357/abf560](https://doi.org/10.3847/1538-4357/abf560).
- 2021 Joshua Krissansen-Totton, Jonathan Fortney, Francis Nimmo, and **Nicholas Wogan**. Oxygen false positives on habitable zone planets around Sun-like stars. *AGU Advances*. DOI: [10.1029/2020AV000294](https://doi.org/10.1029/2020AV000294).
- 2020 **Nicholas Wogan**, Joshua Krissansen-Totton and David Catling. Abundant atmospheric methane from volcanism on terrestrial planets is unlikely and strengthens the case for methane as a biosignature. *Planetary Science Journal*. DOI: [10.3847/PSJ/abb99e](https://doi.org/10.3847/PSJ/abb99e).
- 2020 Kevin Zahnle, Roxana Lupu, David Catling, and **Nicholas Wogan**. Creation and evolution of impact-generated reduced atmospheres of early Earth. *Planetary Science Journal*. DOI: [10.3847/PSJ/ab7e2c](https://doi.org/10.3847/PSJ/ab7e2c).
- 2020 **Nicholas Wogan** and David Catling. When is chemical disequilibrium in Earth-like planetary atmospheres a biosignature versus an anti-biosignature? Disequilibria from dead to living worlds. *Astrophysical Journal*. DOI: [10.3847/1538-4357/ab7b81](https://doi.org/10.3847/1538-4357/ab7b81).

Non-Peer-Reviewed Publications

- 2025 **Nicholas Wogan**, James Mang, Natasha Batalha et al. (2025). The Sonora Substellar Atmosphere Models. V. A Correction to the Disequilibrium Abundance of CO₂ for Sonora Elf Owl. *RNAAS*. DOI: [10.3847/2515-5172/add407](https://doi.org/10.3847/2515-5172/add407).
- 2025 Sukrit Ranjan, ..., **Nicholas Wogan** et al. (2025). Prebiosignatures with the Habitable Worlds Observatory (HWO). *HWO Science Case Development Document*. DOI: [10.48550/arXiv.2507.00165](https://doi.org/10.48550/arXiv.2507.00165).
- 2025 Sukrit Ranjan, ..., **Nicholas Wogan** et al. (2025). Testing Origin-of-Life Theories with the Habitable Worlds Observatory (HWO). *HWO Science Case Development Document*. DOI: [10.48550/arXiv.2507.00164](https://doi.org/10.48550/arXiv.2507.00164).
- 2025 Theresa Fisher, ..., **Nicholas Wogan** et al. (2025). Network and Kinetics-based Biosignatures: Implications for the Putative Habitable World Observatory Design. *HWO Science Case Development Document*. DOI: [10.48550/arXiv.2501.04737](https://doi.org/10.48550/arXiv.2501.04737).
- 2024 Ella Sciamma-O'Brien, Thomas Drant and **Nicholas Wogan**. In an exoplanet atmosphere far, far away. *Nature News & Views*. DOI: [10.1038/s41570-024-00586-2](https://doi.org/10.1038/s41570-024-00586-2).

Research and Observing Proposals Awarded

- 2025 - 2026 Title: "The First Set of Documentation and Tutorials for Any Photochemical Exoplanet Code"
PI: Nicholas Wogan
Program: NASA NNH24ZDA001N-SOSS:F.8 Supplements for Open-Source Science
Total budget: \$50,000
- 2025 - 2028 Title: "Exploring the Influence of Hazes on Climate and Spectra of Small Planets around FGK Stars"
PI: Ravi Kopparapu (NASA Goddard Spaceflight Center)
Program: NASA NNH24ZDA001N-HW:F.4 Habitable Worlds
Total budget: \$484,053
- 2025 - 2028 Title: "The Other Extreme: Enabling Characterization of Metal-poor Brown Dwarfs and testing our Understanding of Jupiter and Saturn"
PI: Sagnick Mukherjee (Arizona State University)
Program: JWST cycle 4 GO, Exoplanets and Exoplanet Formation
Total budget: Not yet allocated.
- 2025 - 2028 Title: "Cliff Hangers: Testing for Atmosphere-Mantle Interactions in Radius Cliff Planets"
PI: Natasha Batalha
Program: JWST cycle 4 GO, Exoplanets and Exoplanet Formation
Total budget: Not yet allocated.

Selected Presentations

Invited

- 2025 **Nicholas Wogan**. The photochemistry and climate of foreign atmospheres: Implications for the origin of life on early earth and understanding exoplanets observed by the James Webb Space Telescope. Planetary Science Seminar, California Insti-

- tute of Technology.
- 2024 **Nicholas Wogan.** Workshop on the `Photochem` code for simulating exoplanet atmospheres. Theoretical Astrophysics Program Planet Formation Initiative Lectureship, University of Arizona.
- 2024 **Nicholas Wogan.** Assessing exoplanet habitability with JWST and the Habitable Worlds Observatory: from the sub-Neptune K2-18b to rocky Earth-like planets. Planetary lunch, University of California Santa Cruz.
- 2024 **Nicholas Wogan.** The photochemistry of hydrogen-rich atmospheres: From the origin of life on Earth to biosignatures on the K2-18b exoplanet. Institute for Geophysics and Planetary Physics seminar, University of California Santa Cruz.
- 2022 **Nicholas Wogan**, David Catling, Kevin Zahnle, and Mark Claire. Rapid timescale for an oxic transition during the Great Oxidation Event and the instability of low atmospheric O₂. NASA Goddard Exoplanets Seminar.
- 2020 **Nicholas Wogan** and David Catling. Atmospheric synthesis of prebiotic molecules on the Hadean Earth. Prebiotic Chemistry and Early Earth Environments Consortium (PCE3), remote conference.
- Contributed
- 2025 **Nicholas Wogan**, Natasha Batalha, Kevin Zahnle, Joshua Krissansen-Totton. Towards a universal chemical and climate model of planetary atmospheres for interpreting HWO observations. Towards the Habitable Worlds Observatory: Visionary Science and Transformational Technology, Washington, D.C.
- 2025 **Nicholas Wogan**, Natasha Batalha, Kevin Zahnle, Joshua Krissansen-Totton. Towards a general chemical and climate model of planetary atmospheres for interpreting exoplanet observations. Exoclines, Montreal, Canada.
- 2025 **Nicholas Wogan**, Natasha Batalha, Joshua Krissansen-Totton, Kevin Zahnle. On inferring the surface production rate of biogenic gases on rocky exoplanets from telescope spectra. American Astronomical Society Meeting, National Harbor, MD.
- 2024 **Nicholas Wogan**, Natasha Batalha, Joshua Krissansen-Totton, Kevin Zahnle. A self-consistent photochemical-climate-flux retrieval method for exoplanet characterization and life detection. American Geophysical Union Fall Meeting, Washington, D.C.
- 2024 **Nicholas Wogan**, Natasha Batalha, Kevin Zahnle, Joshua Krissansen-Totton, Shang-Min Tsai and Renyu Hu. JWST observations of K2-18b can be explained by a gas-rich mini-Neptune with no habitable surface. AbSciCon, Providence, RI.
- 2024 **Nicholas Wogan**, Natasha Batalha, Kevin Zahnle, Joshua Krissansen-Totton and Shang-Min Tsai. JWST observations of K2-18b can be explained by a gas-rich mini-Neptune composition. Extreme Solar Systems V, Christchurch, New Zealand.
- 2023 **Nicholas Wogan**, David Catling, Kevin Zahnle, and Roxana Lupu. Origin of life molecules in the atmosphere after big impacts on the early Earth. American Geophysical Union Fall Meeting, San Francisco, CA.
- 2023 **Nicholas Wogan**, Shawn Domagal-Goldman, Chris Stark, Aki Roberge, Giada Arney, Tyler Robinson. Detecting exo-Earths with the Habitable Worlds Observatory. Habitable Worlds Observatory Conference, Baltimore, MD.

- 2023 **Nicholas Wogan**, David Catling, Kevin Zahnle, and Mark Claire. Rapid timescale for an oxic transition during the Great Oxidation Event and the instability of low atmospheric O₂. Green Bank Astrobiology Conference, Green Bank, WV.
- 2022 **Nicholas Wogan**, David Catling and Kevin Zahnle. Atmospheric nitriles for the origin of life from the atmosphere after large asteroid impacts on the Hadean Earth. Latsis Conference, Zurich, Switzerland.
- 2022 **Nicholas Wogan**, David Catling and Kevin Zahnle. Origin of life chemistry in the atmosphere after large impacts on the early Earth. Astrobiology Science Conference, Atlanta, GA.
- 2021 **Nicholas Wogan**, David Catling and Kevin Zahnle. Molecules for the origin of life from impact-generated atmospheres on early Earth. Simons Foundation Collaboration on the Origin of Life Annual meeting, remote conference.
- 2021 **Nicholas Wogan**, David Catling and Kevin Zahnle. Molecules for the origin of life from impact-generated atmospheres on early Earth. Goldschmidt, remote conference.
- 2019 **Nicholas Wogan** and David Catling. When is chemical disequilibrium in Earth-like planetary atmospheres a biosignature versus an anti-biosignature? Investigating disequilibria from prebiotic to post-biotic worlds. American Geophysical Union Fall Meeting, San Francisco, CA.

Recent Selected Press

- 2025 National Geographic: [Is there really alien life on this exoplanet? We asked 10 experts.](#)
- 2024 New Scientist: [Habitable ocean world K2-18b may actually be inhospitable gas planet](#)
- 2024 YouTube, Dr. Becky: [K2-18b: did JWST really find evidence of life on this exoplanet?](#)
- 2023 Washington Post: [Giant space rock made Earth's ocean boil but also helped early life](#)

Open Source Software

- Photochem: <https://github.com/Nicholaswogan/photochem>. A 1-D photochemical and climate model of planetary atmospheres.
- fortran-yaml-c: <https://github.com/Nicholaswogan/fortran-yaml-c>. A YAML parser and emitter for Fortran.
- Differentia: <https://github.com/Nicholaswogan/Differentia>. Forward mode automatic differentiation for Fortran.
- numbalsoda: <https://github.com/Nicholaswogan/numbalsoda>. A high performance ordinary differential equation solver for Python.